

77.

Să se arate că

$$a) K_m(f)(x) = (m+1) \cdot \sum_{K=0}^m b_{m,K}(x) \int_{\frac{K}{m+1}}^{\frac{K+1}{m+1}} f(t) dt$$

$$b) \Delta_m(f)(x) = (m+1) \cdot \sum_{K=0}^m b_{m,K}(x) \int_0^1 b_{m,K}(t) f(t) dt$$

păstrează aria, unde $K_m: C[0,1] \rightarrow \Pi_m$
 $\Delta_m: C[0,1] \rightarrow \Pi_m$

Rezolvare

a) Mai întâi vom calcula

$$\int_0^1 b_{m,K}(x) dx = \int_0^1 \binom{m}{K} x^K (1-x)^{m-K} dx$$

$$= \int_0^1 x^K (1-x)^{m-K} dx \binom{m}{K}$$

$$= \frac{m!}{K!(m-K)!} B(K+1, m-K+1)$$

$$= \frac{m!}{K!(m-K)!} \cdot \frac{\Gamma(K+1) \cdot \Gamma(m-K+1)}{\Gamma(m+2)}$$

$$= \frac{m!}{K!(m-K)!} \cdot \frac{K!(m-K)!}{(m+1)!} = \frac{1}{m+1}$$

$$\Rightarrow \int_0^1 K_m(f)(x) dx = \int_0^1 \left((m+1) \sum_{K=0}^m b_{m,K}(x) \int_{\frac{K}{m+1}}^{\frac{K+1}{m+1}} f(t) dt \right) dx$$

$$= (m+1) \cdot \sum_{K=0}^m \int_0^1 b_{m,K}(x) dx \int_{\frac{K}{m+1}}^{\frac{K+1}{m+1}} f(t) dt$$

$$= (m+1) \sum_{K=0}^m \frac{1}{m+1} \int_{\frac{K}{m+1}}^{\frac{K+1}{m+1}} f(t) dt$$

$$= \sum_{K=0}^m \int_{\frac{K}{m+1}}^{\frac{K+1}{m+1}} f(t) dt$$

$$= \int_0^1 f(x) dx$$

$$b) \int_0^1 \Delta_m(f)(x) dx =$$

$$= \int_0^1 \left((m+1) \sum_{K=0}^m b_{m,K}(x) \int_0^1 b_{m,K}(t) f(t) dt \right) dx$$

$$= (m+1) \sum_{K=0}^m \int_0^1 b_{m,K}(x) dx \int_0^1 b_{m,K}(t) f(t) dt$$

$$= \sum_{K=0}^m \int_0^1 b_{m,K}(t) f(t) dt$$

$$= \int_0^1 f(t) \cdot \sum_{K=0}^m b_{m,K}(t) dt$$

$$= \int_0^1 f(t) dt$$